

Task 3 Report - MNIST CNN

Introduction:

This report summarizes the CNN model trained on MNIST dataset.

Preprocessing:

- Normalization
- Reshaping
- One-hot encoding

Model Architecture:

- Conv2D layers
- MaxPooling
- Dropout
- Dense output layer

Results:

- Training Accuracy: 0.9978039264678955
- Validation Accuracy: 0.9950000047683716
- Test Accuracy: 0.9943000078201294

Deliverables:

- model.pkl
- confusion_matrix.png
- mnist_samples.png
- cnn_training_history.png

- mnist_errors.png